Peethani Satya Durga Rao

AI Researcher | Machine Learning Engineer | Computer Vision Engineer

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Portfolio Andhra Pradesh, India

Professional Summary

AI Researcher and Software Developer with a strong foundation in Computer Vision, Deep Learning, Machine Learning, and Full-Stack AI System Development. Experienced in designing and developing intelligent software systems that integrate biometric identity management, facial recognition, automated attendance, and enterprise-scale AI applications.

Completed an ANRF-PAIR Research Internship at IIT Hyderabad, conducting research on improving the efficiency of video-based face recognition through tracking-assisted similarity suppression. Passionate about bridging research and real-world deployment by building scalable, production-ready AI systems using modern software engineering practices. Interested in developing scalable AI systems and contributing to research in Computer Vision, Machine Learning, and Intelligent Software Engineering.

Education

Central University of Andhra Pradesh (CUAP) Anantapur, Andhra Pradesh
Master of Science (M.Sc.) in Artificial Intelligence & Data Science *2024 – 2026*

CGPA: **7.60/10.00**

Sri Vasavi Kanyaka Parameshwari & Dr K.S Raju Arts & Science College Penugonda, Andhra Pradesh
Bachelor of Science (B.Sc.) in MCCS *2021 – 2024*

CGPA/Percentage: **8.67**

Sri Vasavi Kanyaka Parameswari & Pithani Venkanna Junior College Penugonda, Andhra Pradesh
Intermediate (MPC) *2019 – 2021*

Percentage: **87.9%**

Zilla Parishad High School Eletipadu, Andhra Pradesh
Secondary School Certificate (SSC) *2018 – 2019*

GPA/Percentage: **8.3**

Research Experience

Master's Dissertation Research Jan 2026 – May 2026
Central University of Andhra Pradesh *Anantapur, Andhra Pradesh*

Research Topic: A Scalable and Efficient Similarity Search Framework for Large-Scale Face Recognition Systems

- Conducted research on scalable face recognition and similarity search techniques for biometric identification systems.
- Investigated the performance, scalability, and retrieval efficiency of different similarity search approaches through comparative experimental analysis.
- Performed literature review, research gap identification, experimental design, implementation, and result evaluation.

DOI/URL: ijirt.org/publishedpaper/IJIRT201256_PAPER.pdf

ANRF-PAIR Research Intern May 2026 – Jul 2026
Indian Institute of Technology Hyderabad (IIT Hyderabad) *Hyderabad, Telangana*

Research Topic: Efficient Face Recognition in Video Streams using Tracking-Assisted Similarity Suppression

- Investigated computational challenges in real-time video-based face recognition through an extensive review of state-of-the-art methods.
- Identified research gaps related to redundant face recognition during continuous video processing and explored tracking-assisted recognition strategies.
- Proposed an efficient recognition framework aimed at reducing unnecessary similarity search operations while maintaining recognition performance.

Projects

BioVision Enterprise **(2026 – Present)**
github.com/satya24mai26-gif/biovision-enterprise

AI-powered campus identity platform integrating biometric identity management, intelligent attendance, surveillance, and administrative services.

- Designed a scalable enterprise architecture for AI-based campus identity management.
- Integrated face recognition, attendance automation, QR authentication, and multi-camera support.
- Developed modular backend and frontend components for future extensibility.

Tech: Python | FastAPI | React | Next.js | OpenCV | InsightFace | FAISS | MongoDB

Smart Attendance System **(2025)**
github.com/satya24mai26-gif/smart-facial-attendance

AI-based attendance management system developed during the M.Sc. internship.

- Automated attendance using facial recognition and biometric authentication.
- Implemented face enrollment, recognition, attendance recording, and reporting.
- Integrated efficient facial embedding retrieval and database management.

Tech: Python | OpenCV | FaceNet | FAISS | SQLite

ProjectMind AI **(2026)**
github.com/satya24mai26-gif/ProjectMind-AI

AI-powered knowledge graph platform for intelligent project and research management.

- Developed interactive knowledge graph visualization and management.
- Integrated LLM-based AI assistance for summaries and relationship generation.
- Built scalable full-stack architecture using FastAPI and Next.js.

Tech: Next.js | React | FastAPI | SQLAlchemy | Ollama | SQLite

Cinematic Invitation SaaS Platform **(2024)**
github.com/satya24mai26-gif/cinematic-invitation-platform

cinematic-invitation-platform.vercel.app

Full-stack SaaS platform for creating customizable digital invitation websites.

- Developed responsive invitation templates with dynamic content management.
- Implemented secure authentication, dashboard management, and REST APIs.
- Built scalable frontend and backend using the MERN ecosystem.

Tech: React | Node.js | Express.js | MongoDB | Tailwind CSS

Technical Skills

Programming: Python | JavaScript | TypeScript | SQL

AI & Computer Vision: Machine Learning | Deep Learning | Computer Vision | Face Recognition | Similarity Search | LLMs | RAG

Frameworks & Libraries: PyTorch | TensorFlow | OpenCV | InsightFace | ArcFace | FaceNet | FAISS | NumPy | Pandas | Scikit-learn

Full-Stack Development: FastAPI | React | Next.js | Node.js | Express.js | REST APIs | Tailwind CSS

Databases: MongoDB | SQLite

Tools: Git | GitHub | Linux | VS Code | LaTeX | Jupyter Notebook | Postman

Earlier Experience

Python Full-Stack Intern 2023 – 2024
Brain of Vision

- Completed training in Python, HTML, CSS, JavaScript, and full-stack web development.
- Developed foundational skills in frontend, backend, and database-driven web applications.

Community Survey Intern 2022 – 2023
Public Health Awareness Survey

- Conducted field surveys to assess public awareness of chronic diseases and collected community data for analysis.

Industrial Intern 2022 – 2023
Lakshmi Plastic Industry, Marteru

- Observed polymer-based plastic bag manufacturing processes, production workflow, and quality control practices.

Achievements

- Selected for the prestigious ANRF-PAIR Research Internship at Indian Institute of Technology Hyderabad.
- Developed multiple end-to-end Artificial Intelligence applications integrating Computer Vision, Full-Stack Development, and Enterprise Software Engineering.
- Designed scalable AI system architectures for intelligent biometric identity management and smart campus infrastructure.
- Built research-oriented software platforms bridging academic research with practical AI deployment.